

Name: _____
Class: _____
Date: _____

Mr. Rawson • IB DP Physics

Simple Harmonic Motion 7 – Practice IB Questions

Answer Key

Before you start

Have your **calculator in radians** and your **IB data booklet** open. In every calculation, write the equation first, then substitute, then give the answer with a unit. **Check:** $\sin(\pi/2) = 1$ on your calculator.

The command terms on this sheet, as the IB defines them

State	Give a specific name, value or other brief answer without explanation or calculation.
Calculate	Obtain a numerical answer showing the relevant stages in the working.
Show	Give the steps in a calculation or derivation. In <i>show that</i> , the answer is given: the marks are for the working, so give one more significant figure than the value shown.
Determine	Obtain the only possible answer.
Explain	Give a detailed account including reasons or causes.

SL

Everyone answers this section.

Multiple choice (Paper 1A style: circle one letter)

Question 1

1 mark

Relevant equations: $\omega = 2\pi f$

An object oscillates with simple harmonic motion at a frequency of 2.5 Hz. What is the angular frequency of the oscillation?

- A. 0.40 rad/s B. 2.5 rad/s C. 7.9 rad/s D. 16 rad/s

Answer: D $\omega = 2\pi f = 2\pi \times 2.5 = 15.7$. A is T , B is f , C is πf .

Question 2

1 mark

Relevant equations: $T = 2\pi\sqrt{\frac{m}{k}}$ $v_{\max} = \omega x_0$ $a_{\max} = \omega^2 x_0$

A mass on a spring oscillates with simple harmonic motion. The amplitude is doubled. What happens to the period T , the maximum speed $v_{\max}$ and the maximum acceleration $a_{\max}$?

	T	$v_{\max}$	$a_{\max}$
A.	unchanged	unchanged	unchanged
B.	unchanged	doubled	doubled
C.	doubled	unchanged	unchanged
D.	doubled	doubled	quadrupled

Answer: B ω does not depend on amplitude, so T is unchanged; $v_{\max} = \omega x_0$ and $a_{\max} = \omega^2 x_0$ both scale with x_0 .

Question 3

1 mark

Relevant equations: $v_{\max} = \omega x_0$ $a_{\max} = \omega^2 x_0$ $T = \frac{2\pi}{\omega}$

An oscillator in simple harmonic motion has a maximum speed of 0.60 m/s and a maximum acceleration of 3.0 m/s². What is its period?

- A. 0.20 s B. 1.3 s C. 5.0 s D. 7.9 s

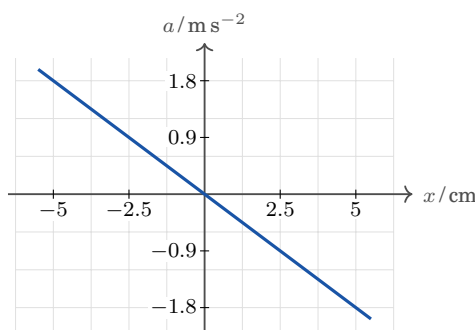
Answer: B $a_{\max}/v_{\max} = \omega^2 x_0/\omega x_0 = \omega = 5.0$, so $T = 2\pi/5.0 = 1.26$. A is $1/\omega$; C is ω .

Question 4

1 mark

Relevant equations: $a = -\omega^2 x$ $\omega = 2\pi f$

The graph shows how the acceleration a of an oscillating object varies with its displacement x .



What is the frequency of the oscillation?

- A. 0.95 Hz B. 5.7 Hz C. 6.0 Hz D. 36 Hz

Answer: A gradient = $-1.8/0.050 = -36$, so $\omega^2 = 36$, $\omega = 6.0$ and $f = \omega/2\pi = 0.95$. C is ω ; D is the gradient; B is $36/2\pi$.

Structured questions (Paper 2 style)

Question 5

6 marks

Relevant equations: $\omega = \frac{2\pi}{T}$ $v_{\max} = \omega x_0$ $a_{\max} = \omega^2 x_0$ $a = -\omega^2 x$

A car drives over a bump. Afterwards, the car body moves up and down on its springs with simple harmonic motion of period 0.80 s and amplitude 4.0 cm.

- (a) Calculate the angular frequency of the oscillation.

[1]

Mark scheme

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{0.80} = 7.9 \text{ «rad s}^{-1}\text{» } \checkmark$$

- (b) Calculate the maximum speed of the car body.

[1]

Mark scheme

$$v_{\max} = \omega x_0 = 7.85 \times 0.040 = 0.31 \text{ «m s}^{-1}\text{» } \checkmark$$

Allow ECF from (a). Award [0] for $7.85 \times 4.0 = 31$: the amplitude must be in metres.

- (c) Calculate the maximum acceleration of the car body, and state where in the oscillation it occurs.

[2]

Mark scheme

$$a_{\max} = \omega^2 x_0 = 7.85^2 \times 0.040 = 2.5 \text{ «m s}^{-2}\text{» } \checkmark$$

at maximum displacement / at either extreme of the motion / at $x = \pm x_0$ ✓

- (d) The motion of the car body is described by $a = -\omega^2 x$. State **two** features of the acceleration that this equation describes.

[2]

Mark scheme

acceleration is proportional to displacement «from equilibrium» ✓

acceleration is directed towards equilibrium / opposite to the displacement ✓

Do not accept “the acceleration is negative” for the second mark.

Question 6

6 marks

Relevant equations: $T = 2\pi\sqrt{\frac{l}{g}}$ $\omega = \frac{2\pi}{T}$ $v_{\max} = \omega x_0$

A child sits on a playground swing. The swing behaves as a simple pendulum of length 2.5 m.

- (a) Show that the period of the swing is about 3.2 s.

[2]

Mark scheme

$$T = 2\pi\sqrt{\frac{l}{g}} = 2\pi\sqrt{\frac{2.5}{9.81}} \checkmark$$

$$= 3.17 \text{ «s» } \checkmark$$

Answer must be given to at least 3 s.f., or the working must be complete.

- (b) The seat moves 0.60 m either side of its equilibrium position. Calculate the maximum speed of the seat.

[2]

Mark scheme

$$\omega = \frac{2\pi}{3.17} = 1.98 \text{ «rad s}^{-1}\text{» } \checkmark$$

$$v_{\max} = \omega x_0 = 1.98 \times 0.60 = 1.2 \text{ «m s}^{-1}\text{» } \checkmark$$

Allow use of $T = 3.2 \text{ s}$, giving 1.18 m s^{-1} .

- (c) A heavier child sits on the same swing. State and explain the effect on the period.

[2]

Mark scheme

no change / period stays the same ✓

$T = 2\pi\sqrt{l/g}$ does not contain the mass / T depends only on l and g ✓

Question 7

4 marks

Relevant equations: $\omega = \sqrt{\frac{k}{m}}$ $T = \frac{2\pi}{\omega}$ $a = -\omega^2 x$

A trolley of mass 0.40 kg sits between two springs on a level track. When it is displaced by x from its equilibrium position, the resultant force on it is $F = -36x$, with F in newtons and x in metres.

- (a) Explain why the trolley performs simple harmonic motion when it is released.

[2]

Mark scheme

the «resultant» force/acceleration is proportional to the displacement ✓

and directed towards equilibrium / opposite to the displacement «shown by the minus sign» ✓

- (b) Calculate the period of the oscillation.

[2]

Mark scheme

$$\omega = \sqrt{\frac{36}{0.40}} = 9.5 \text{ «rad s}^{-1}\text{» OR } T = 2\pi\sqrt{\frac{0.40}{36}} \checkmark$$

$$T = 0.66 \text{ «s» } \checkmark$$

Award [2] for a bald correct answer.

HL students: answer this section as well. It uses the HL equations from your data booklet.

Multiple choice (Paper 1A style: circle one letter)

Question 8

1 mark

Relevant equations: $x = x_0 \sin \omega t$ $\omega = \frac{2\pi}{T}$

An object moves in simple harmonic motion with period T and amplitude x_0 . It passes through the equilibrium position at $t = 0$. What is the shortest time for it to reach a displacement of $\frac{1}{2}x_0$?

- A. $\frac{T}{12}$ B. $\frac{T}{8}$ C. $\frac{T}{6}$ D. $\frac{T}{4}$

Answer: A $\sin \omega t = \frac{1}{2}$ gives $\omega t = \pi/6$, and $\pi/6$ is one twelfth of 2π . B assumes the displacement grows steadily with time, which it does not.

Question 9

1 mark

Relevant equations: $x = x_0 \sin(\omega t + \phi)$ $v = \omega x_0 \cos(\omega t + \phi)$

The displacement of an oscillator is given by $x = x_0 \sin(\omega t + \phi)$. At $t = 0$ the oscillator passes through the equilibrium position moving in the negative direction. What is ϕ ?

- A. 0 B. $\frac{\pi}{2}$ C. π D. $\frac{3\pi}{2}$

Answer: C $\sin \pi = 0$, and $v = \omega x_0 \cos \pi = -\omega x_0$, negative. A also gives $x = 0$, but moving in the positive direction.

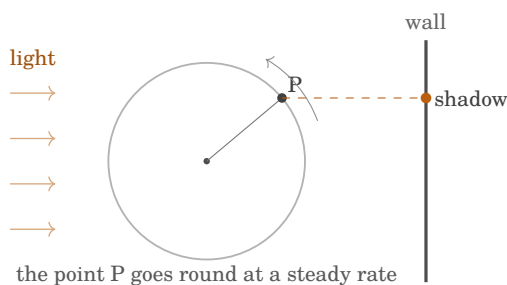
Structured questions (Paper 2 style)

Question 10

7 marks

Relevant equations: $\omega = 2\pi f$ $x = x_0 \sin \omega t$ $v = \omega x_0 \cos \omega t$ $v = \pm \omega \sqrt{x_0^2 - x^2}$

A small peg P is fixed to the edge of a turntable, 0.12 m from its centre. The turntable rotates at a steady 45 revolutions per minute. Light shines from the side, so the shadow of P on a wall moves up and down with simple harmonic motion.



- (a) Show that the angular frequency of the shadow's motion is about 4.7 rad/s.

[1]

Mark scheme

$$\omega = 2\pi f = 2\pi \times \frac{45}{60} = 4.71 \text{ «rad s}^{-1}\text{» } \checkmark$$

- (b) At $t = 0$ the shadow passes through its equilibrium position, moving upwards. Calculate the displacement of the shadow at $t = 0.20$ s.

[2]

Mark scheme

$$\omega t = 4.71 \times 0.20 = 0.942 \text{ «rad» } \checkmark$$

$$x = x_0 \sin \omega t = 0.12 \sin(0.942) = 0.097 \text{ «m» } \checkmark$$

Calculator in degrees gives 0.0020 m: award [1] max.

- (c) Calculate the velocity of the shadow at $t = 0.20$ s.

[2]

Mark scheme

$$v = \omega x_0 \cos \omega t \checkmark$$

$$= 4.71 \times 0.12 \times \cos(0.942) = 0.33 \text{ «m s}^{-1}\text{», upwards } \checkmark$$

Allow ECF from (b).

- (d) Use your answer to (b) to confirm your answer to (c).

[1]

Mark scheme

$$v = \omega \sqrt{x_0^2 - x^2} = 4.71 \sqrt{0.12^2 - 0.097^2} = 0.33 \text{ «m s}^{-1}\text{» } \checkmark$$

- (e) A second peg is fixed to the edge of the same turntable, a quarter of a turn ahead of P. State the phase difference between the motions of the two shadows.

[1]

Mark scheme

$$\frac{\pi}{2} \text{ «rad» } \checkmark$$

Allow 90° .